

(*Didinium -- Flow Field*)

In[1]:= (*Parameters*)

```

n = 12;
ρvar = 1.3;
θvar = 30 π / 180;
φvar = 14 π / 180;
radius = ρvar Sin[θvar];
n1 = 24;
n2 = 48;
ρ2var = 1.3;
θ2var = 100 π / 180;
radius2 = ρ2var Sin[θ2var];
arrowLength = 0.2;
shaftRadius = 0.01;
headLength = 0.1;
headRadius = 0.05;

```

Gbulk[Xx_ : {x, y, z}, Xy_ : {xs, ys, zs}] :=

```

Module[{RX = Xx - Xy, δ = IdentityMatrix[3]},
Table[ $\frac{\delta_{[j], [k]}}{\sqrt{RX.RX}} + \frac{RX_{[j]} RX_{[k]}}{(RX.RX)^{3/2}}$ , {j, 1, 3}, {k, 1, 3}]]

```

GbulkLap[{x_, y_, z_}, {xs_, ys_, zs_}] := Evaluate[

```
FullSimplify[Laplacian[Gbulk[{x, y, z}, {xs, ys, zs}][All, 3], {xs, ys, zs}]]]

```

Gsphere[Xx_ : {x, y, z}, Xy_ : {xf, yf, zf}, XyB_ : {xs, ys, zs}, A_ : 1] :=

```

Module[{Xys =  $\frac{A^2}{(Xy.Xy)}$  Xy, RX = Xx - Xy - XyB,
RXs = Xx -  $\frac{A^2}{(Xy.Xy)}$  Xy - XyB, rX =  $\sqrt{(Xx - Xy - XyB).(Xx - Xy) - XyB}$ ,
rXs =  $\sqrt{\left(Xx - \frac{A^2}{(Xy.Xy)} Xy - XyB\right).\left(Xx - \frac{A^2}{(Xy.Xy)} Xy - XyB\right)}$ ,
XyF = Xy, XxB = Xx - XyB, δ = IdentityMatrix[3]}, Table[
-  $\frac{A}{\sqrt{XyF.XyF}} \frac{\delta_{[j], [k]}}{rXs} - \frac{A^3}{(XyF.XyF)^{3/2}} \frac{RXs_{[j]} RXs_{[k]}}{rXs^3} -$ 
 $\frac{XyF.XyF - A^2}{\sqrt{XyF.XyF}} \left( \frac{Xys_{[j]} Xys_{[k]}}{A^3 rXs} - \frac{A}{(XyF.XyF) rXs^3} (Xys_{[j]} RXs_{[k]} + Xys_{[k]} RXs_{[j]}) + \right.$ 

```

$$\begin{aligned}
& \left(\frac{2 Xys[[j]] Xys[[k]] (Xys.RXs)}{A^3 rXs^3} \right) - ((XxB.XxB) - A^2) \left(\frac{(XyF.XyF) - A^2}{2 (XyF.XyF)^{3/2}} \right) \left(\right. \\
& \frac{-3 RXs[[j]] XyF[[k]]}{A rXs^3} + A \frac{\delta[[j, k]]}{rXs^3} - \\
& 3 A \frac{RXs[[j]] RXs[[k]]}{rXs^5} - \frac{2 Xys[[j]] XyF[[k]]}{A rXs^3} + \frac{6 RXs[[j]] XyF[[k]]}{A rXs^5} Xys.RXs + \\
& \frac{3 A}{Xys.Xys} \left((RXs[[j]] Xys[[k]] rXs^2 + RXs[[j]] RXs[[k]] (Xys.Xys) + (rXs - \sqrt{Xys.Xys}) rXs^2 \right. \\
& \left. \sqrt{Xys.Xys} \delta[[j, k]] \right) / \left(rXs^3 (rXs \sqrt{Xys.Xys} + XxB.Xys - Xys.Xys) \right) - \\
& \frac{3 A}{Xys.Xys} \left(\left(\left(\sqrt{Xys.Xys} RXs[[j]] + rXs Xys[[j]] \right) (Xys[[k]] rXs^2 - Xys.Xys RXs[[k]] + \right. \right. \\
& \left. \left. (XxB[[k]] - 2 Xys[[k]]) rXs \sqrt{Xys.Xys} \right) \right) / \\
& \left(rXs^2 (rXs \sqrt{Xys.Xys} + XxB.Xys - Xys.Xys)^2 \right) - \\
& \frac{3 A}{Xys.Xys} \frac{XxB[[j]] Xys[[k]] + \sqrt{XxB.XxB} \sqrt{Xys.Xys} \delta[[j, k]]}{\sqrt{XxB.XxB} (\sqrt{XxB.XxB} \sqrt{Xys.Xys} + XxB.Xys)} + \frac{3 A}{Xys.Xys} \\
& \left. \frac{(\sqrt{Xys.Xys} XxB[[j]] + \sqrt{XxB.XxB} Xys[[j]]) (\sqrt{Xys.Xys} XxB[[k]] + \sqrt{XxB.XxB} Xys[[k]])}{\sqrt{XxB.XxB} (\sqrt{XxB.XxB} \sqrt{Xys.Xys} + XxB.Xys)^2} \right) \\
& , \{j, 1, 3\}, \{k, 1, 3\} \} \\
& \text{uSwimmerN}[\{x_, y_, z_ \}, \{xs_, ys_, zs_ \}, \\
& \{zf_, \rho f_ \}, A_, V_, \Omega_, NN_Integer] := \text{Module}[\\
& \{RXX = \{x, y, z\} - \{xs, ys, zs\}, \\
& Fz = -\frac{1}{NN} \frac{1}{1 - \frac{A^3 (-2 zf^2 + \rho f^2) + 3 A (zf^2 + \rho f^2) (2 zf^2 + \rho f^2)}{4 (zf^2 + \rho f^2)^{5/2}}} \frac{6 A V}{8}, F\phi = \frac{1}{NN} \frac{1}{\left(1 - \left(\frac{A^2}{zf^2 + \rho f^2}\right)^{3/2}\right)} \frac{A^3 \Omega}{\rho f} \}, \\
& \frac{3 A V}{4} \left(\text{Gbulk}[\{x, y, z\}, \{xs, ys, zs\}] \llbracket \text{All}, 3 \rrbracket + \frac{A^2}{6} \text{GbulkLap}[\{x, y, z\}, \{xs, ys, zs\}] \right) \\
& + \frac{A^3}{\text{Norm}[RXX]^3} \text{Cross}[\{0, 0, \Omega\}, RXX] \\
& + \text{Sum} \left[\left(\text{Gbulk}[\{x, y, z\}, \{xs, ys, zs\}] + \left\{ \rho f \cos \left[2 \pi \frac{i-1}{NN} \right], \rho f \sin \left[2 \pi \frac{i-1}{NN} \right], zf \right\} \right) + \right. \\
& \left. \text{Gsphere}[\{x, y, z\}, \left\{ \rho f \cos \left[2 \pi \frac{i-1}{NN} \right], \rho f \sin \left[2 \pi \frac{i-1}{NN} \right], zf \right\}, \{xs, ys, zs\}, A] \right) . \\
& \left\{ F\phi \sin \left[2 \pi \frac{i-1}{NN} \right], -F\phi \cos \left[2 \pi \frac{i-1}{NN} \right], Fz \right\}, \{i, 1, NN\} \\
&]
\end{aligned}$$

(*Force arrow plots*)

(*Straight*)

```

In[19]:= 1 - 
$$\frac{A^3 (-2 z f^2 + \rho f^2) + 3 A (z f^2 + \rho f^2) (2 z f^2 + \rho f^2)}{4 (z f^2 + \rho f^2)^{5/2}}$$
 /.
{zf -> pvar Cos[thvar], pf -> pvar Sin[thvar], zf2 -> p2var Cos[th2var],
 pf2 -> p2var Sin[th2var], A -> 1, V -> 1, NN -> 12, MM -> 12, th -> 0. Pi / 180}
1 - 
$$\frac{A^3 (-2 z f^2 + \rho f^2) + 3 A (z f^2 + \rho f^2) (2 z f^2 + \rho f^2)}{4 (z f^2 + \rho f^2)^{5/2}}$$
 /.
{zf -> pvar Cos[thvar], pf -> pvar Sin[thvar], zf2 -> p2var Cos[th2var],
 pf2 -> p2var Sin[th2var], A -> 1, V -> 1, NN -> 12, MM -> 12, th -> 0. Pi / 180}

Out[19]=
0.132624

Out[20]=
0.302183

```

```

In[21]:= (*Function to create a single tube arrow*)
tubeArrow[base_, dir_] := Module[{shaftEnd, headBase, headTip},
  shaftEnd = base + dir (1 - headLength / arrowLength);
  headBase = shaftEnd;
  headTip = base + dir;
  {Tube[{base, shaftEnd}, shaftRadius], Cone[{headBase, headTip}, headRadius]}]

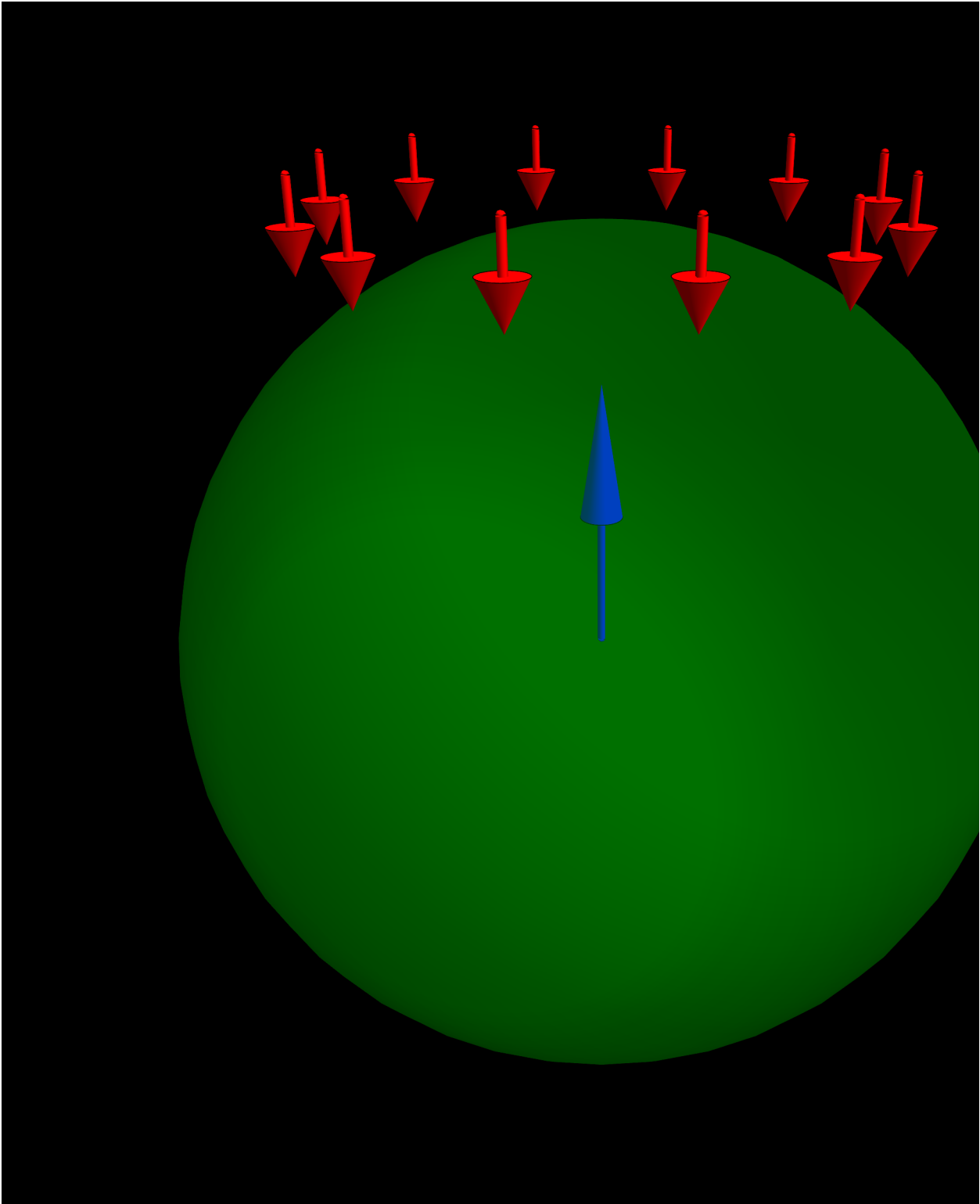
(*Generate arrows*)
arrows = Table[Module[{theta, base, dir}, theta = 2 Pi k / n;
  (*Base point on ring*)
  base = {radius Cos[theta], radius Sin[theta], rho var Cos[theta var]};
  (*z direction*) dir = arrowLength {0, 0, -1};
  (*Tangential direction *) (*dir=arrowLength {-Sin[theta], Cos[theta], 0};*)
  tubeArrow[base, dir]], {k, 0, n - 1}];

axes = Table[Graphics3D[{White, tubeArrow[{1, -1 / 2, -1}, 0.3 k]}],
  {k, {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}}}];

(*Render*)
plotToSave = Show[
  Graphics3D[{Opacity[0.25], Green, Sphere[{0, 0, 0}, 1]}],
  Graphics3D[{Red, arrows}],
  Graphics3D[{Blue, tubeArrow[{0, 0, 0}, {0, 0, 0.65}]}],
  Background → Black, ImageSize → 800 {1, 1}, Boxed → False,
  Axes → False, Lighting → "Neutral", ViewVector → {{2, 2, 1.5}, {0, 0, 0.1}},
  ViewVertical → {0, 0, 1}, ViewAngle → 50 Pi / 180]
SetDirectory[NotebookDirectory[]];
Export["ArrowsA.jpg", plotToSave]

```

Out[24]=



Out[26]=

ArrowsA.jpg

(*Spinner*)

```
In[27]:= 
$$\left(1 - \left(\frac{A^2}{zf^2 + \rho f^2}\right)^{3/2}\right) /. \{zf \rightarrow \rho var \cos[\theta var], \rho f \rightarrow \rho var \sin[\theta var], zf2 \rightarrow \rho2 var \cos[\theta2 var],$$


$$\rho f2 \rightarrow \rho2 var \sin[\theta2 var], A \rightarrow 1, V \rightarrow 1, NN \rightarrow 12, MM \rightarrow 12, \theta \rightarrow 0. \pi / 180\}$$


$$\left(1 - \left(\frac{A^2}{zf2^2 + \rho f2^2}\right)^{3/2}\right) /. \{zf \rightarrow \rho var \cos[\theta var], \rho f \rightarrow \rho var \sin[\theta var], zf2 \rightarrow \rho2 var \cos[100 \pi / 180.],$$


$$\rho f2 \rightarrow \rho2 var \sin[100 \pi / 180.], A \rightarrow 1, V \rightarrow 1, NN \rightarrow 12, MM \rightarrow 12, \theta \rightarrow 0. \pi / 180\}$$

```

```
Out[27]= 0.544834
```

```
Out[28]= 0.544834
```

```
In[29]:= theta 180. /  $\pi$ 
```

```
Out[29]= 30.
```

```

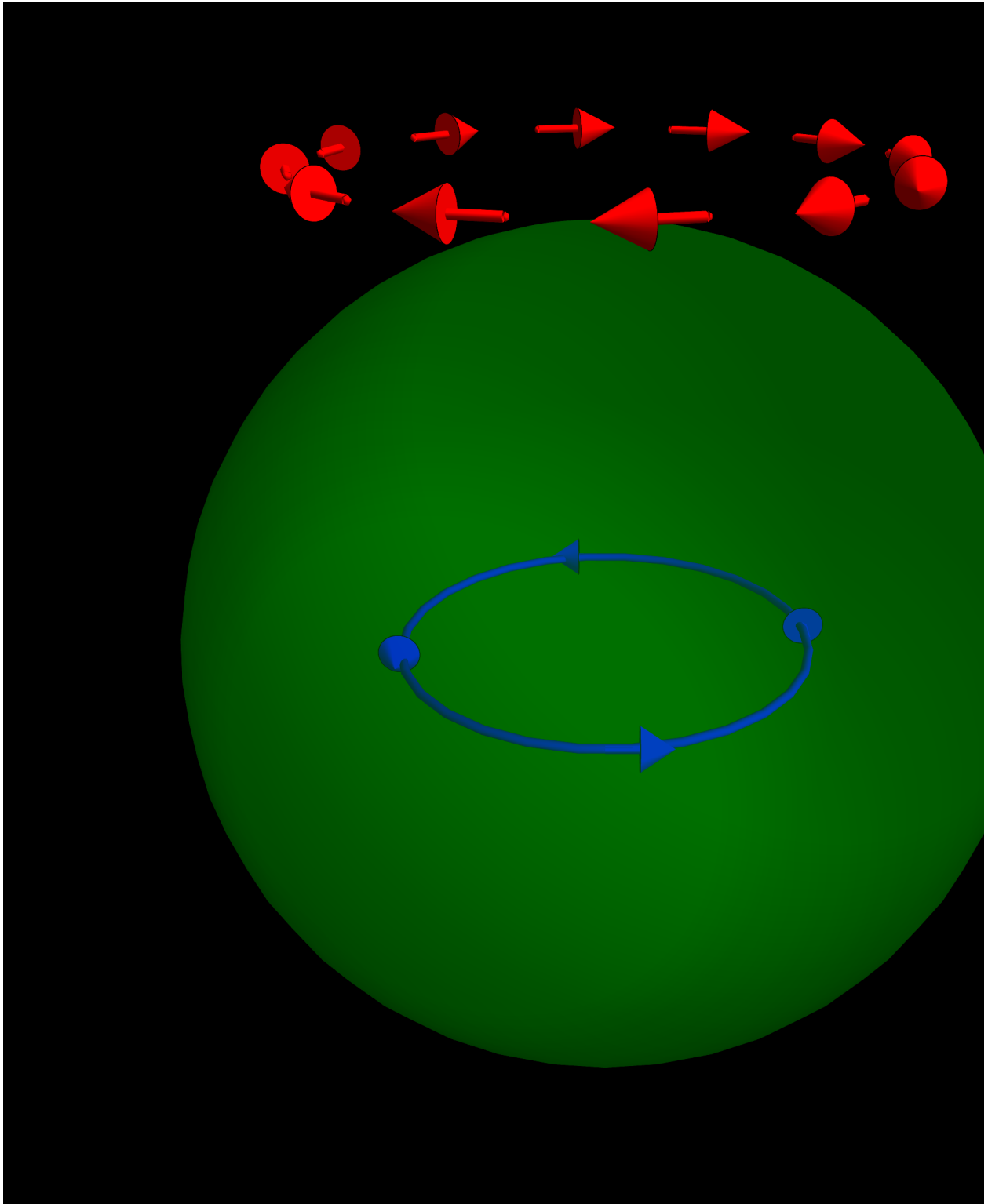
In[30]:= (*Function to create a single tube arrow*)
tubeArrow[base_, dir_] := Module[{shaftEnd, headBase, headTip},
  shaftEnd = base + dir (1 - headLength / arrowLength);
  headBase = shaftEnd;
  headTip = base + dir;
  {Tube[{base, shaftEnd}, shaftRadius], Cone[{headBase, headTip}, headRadius]}]

(*Generate arrows*)
arrows = Table[Module[{theta, base, dir}, theta = 2 Pi k / n;
  (*Base point on ring*)
  base = {radius Cos[theta], radius Sin[theta], rho var Cos[theta]};
  (*Tangential direction *) dir = arrowLength {Sin[theta], -Cos[theta], 0};
  tubeArrow[base, dir]], {k, 0, n - 1}];
arrowsB = Table[Module[{theta, base, dir}, theta = 2 Pi k / 4 + Pi / 4;
  (*Base point on ring*) base = 0.5 {Cos[theta], Sin[theta], 0};
  (*Tangential direction *) dir = 0.15 {-Sin[theta], Cos[theta], 0};
  tubeArrow[base, dir]], {k, 0, 4 - 1}];
ring = ParametricPlot3D[{(0.5 + 0.01 Cos[v]) Cos[u], (0.5 + 0.01 Cos[v]) Sin[u],
  0.01 Sin[v]}, {u, 0, 2 Pi}, {v, 0, 2 Pi}, PlotStyle -> Blue, Mesh -> None];

(*Render*)
plotToSave = Show[
  Graphics3D[{Opacity[0.25], Green, Sphere[{0, 0, 0}, 1]}],
  Graphics3D[{Red, arrows}], Graphics3D[{Blue, arrowsB}], ring,
  Background -> Black, ImageSize -> 800 {1, 1}, Boxed -> False,
  Axes -> False, Lighting -> "Neutral", ViewVector -> {{2, 2, 1.5}, {0, 0, 0.1}},
  ViewVertical -> {0, 0, 1}, ViewAngle -> 50 Pi / 180]
SetDirectory[NotebookDirectory[]];
Export["ArrowsB.jpg", plotToSave]

```

Out[34]=



Out[36]=

ArrowsB.jpg

(*Didinium*)

```

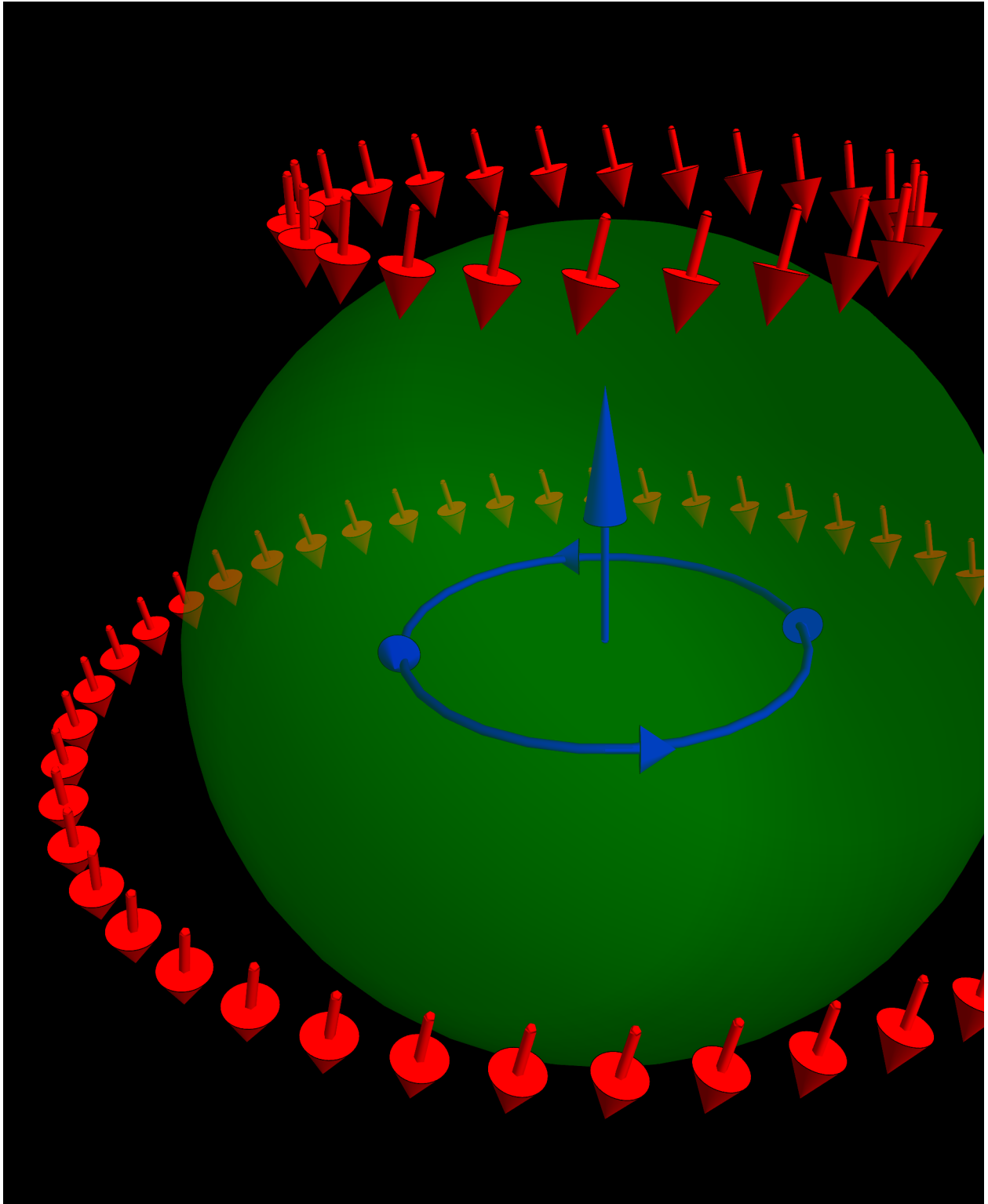
In[37]:= (*Function to create a single tube arrow*)
tubeArrow[base_, dir_] := Module[{shaftEnd, headBase, headTip},
  shaftEnd = base + dir (1 - headLength / arrowLength);
  headBase = shaftEnd;
  headTip = base + dir;
  {Tube[{base, shaftEnd}, shaftRadius], Cone[{headBase, headTip}, headRadius]}]

(*Generate arrows*)
arrows = Table[Module[{theta, base, dir}, theta = 2 Pi k / n1;
  (*Base point on ring*)
  base = {radius Cos[theta], radius Sin[theta], rho var Cos[theta var]};
  (*direction *)
  dir = arrowLength {Sin[theta] x Sin[phi var], -Cos[theta] x Sin[phi var], -Cos[phi var]};
  tubeArrow[base, dir]], {k, 0, n1 - 1}];
arrows2 = Table[Module[{theta, base, dir}, theta = 2 Pi k / n2;
  (*Base point on ring*)
  base = {radius2 Cos[theta], radius2 Sin[theta], rho var Cos[theta2 var]};
  (*direction *)
  dir = arrowLength {Sin[theta] x Sin[phi var], -Cos[theta] x Sin[phi var], -Cos[phi var]};
  tubeArrow[base - dir, dir]], {k, 0, n2 - 1}];

(*Render*)
plotToSave = Show[
  Graphics3D[{Opacity[0.25], Green, Sphere[{0, 0, 0}, 1]}],
  Graphics3D[{Red, arrows}], Graphics3D[{Red, arrows2}],
  Graphics3D[{Blue, tubeArrow[{0, 0, 0}, {0, 0, 0.65}]}],
  Graphics3D[{Blue, arrowsB}], ring,
  Background -> Black, ImageSize -> 800 {1, 1}, Boxed -> False,
  Axes -> False, Lighting -> "Neutral", ViewVector -> {{2, 2, 1.5}, {0, 0, 0.1}},
  ViewVertical -> {0, 0, 1}, ViewAngle -> 50 Pi / 180]
SetDirectory[NotebookDirectory[]]
Export["ArrowsC.jpg", plotToSave]

```

Out[40]=



Out[41]=

/Users/amaths/Library/CloudStorage/Box-Box/Mathijssen/My
Publications/2026-01 Guillermina Didinium/S4

Out[42]=

ArrowsC.jpg

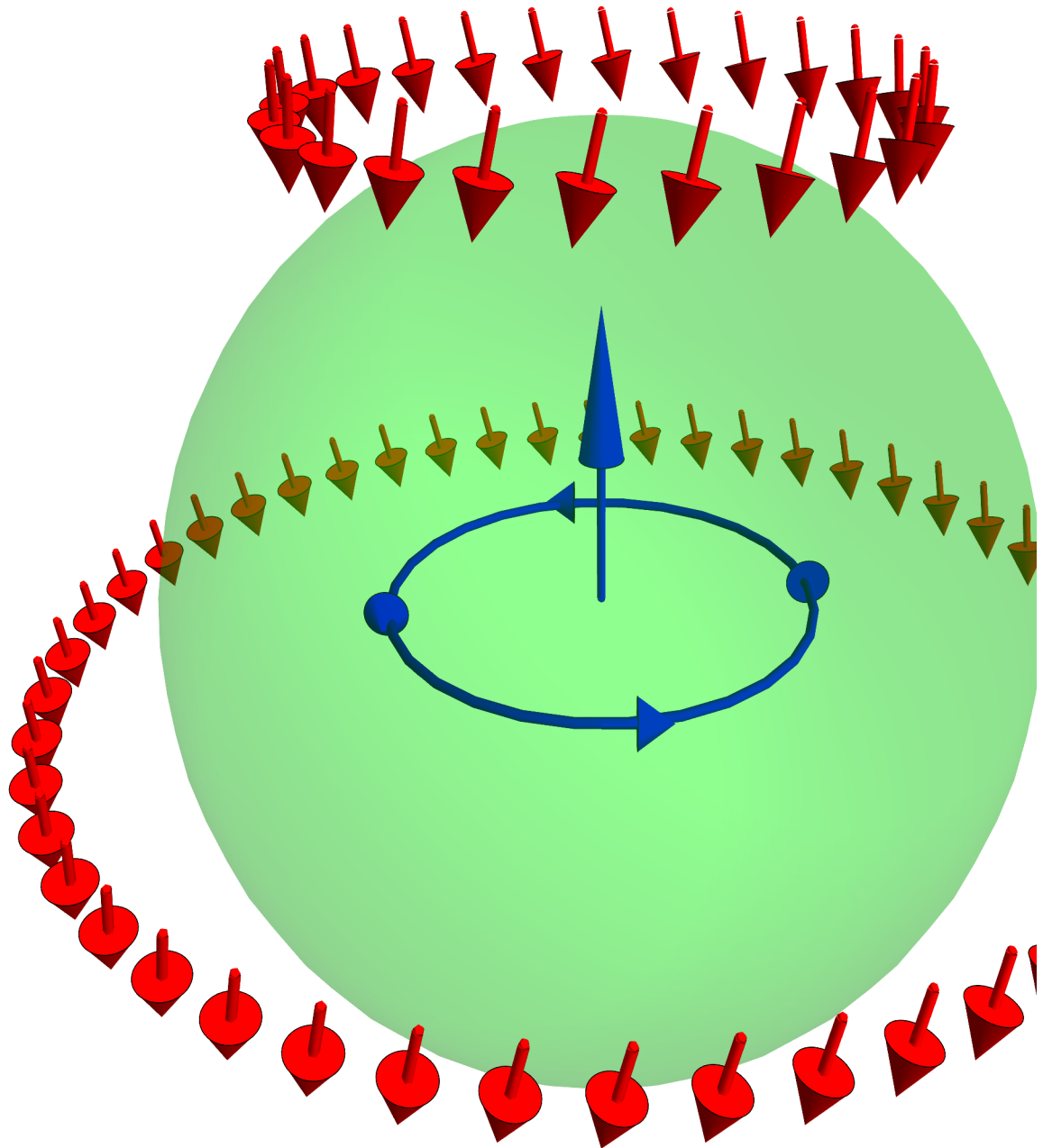
In[43]:=

```

plotToSave = Show[
  Graphics3D[{Opacity[0.25], Green, Sphere[{0, 0, 0}, 1]}],
  Graphics3D[{Red, arrows}], Graphics3D[{Red, arrows2}],
  Graphics3D[{Blue, tubeArrow[{0, 0, 0}, {0, 0, 0.65}]}],
  Graphics3D[{Blue, arrowsB}], ring,
  Background → White, ImageSize → 800 {1, 1}, Boxed → False,
  Axes → False, Lighting → "Neutral", ViewVector → {{2, 2, 1.5}, {0, 0, 0.1}},
  ViewVertical → {0, 0, 1}, ViewAngle → 50  $\pi$  / 180, AspectRatio → 1.1]
SetDirectory[NotebookDirectory[]]
Export["DidiniumModel.jpg", plotToSave]

```

Out[43]=



Out[44]=

/Users/amaths/Library/CloudStorage/Box-Box/Mathijssen/My
Publications/2026-01 Guillermina Didinium/S4

Out[45]=

DidiniumModel.jpg

(*Flows - Straight swimmer*)

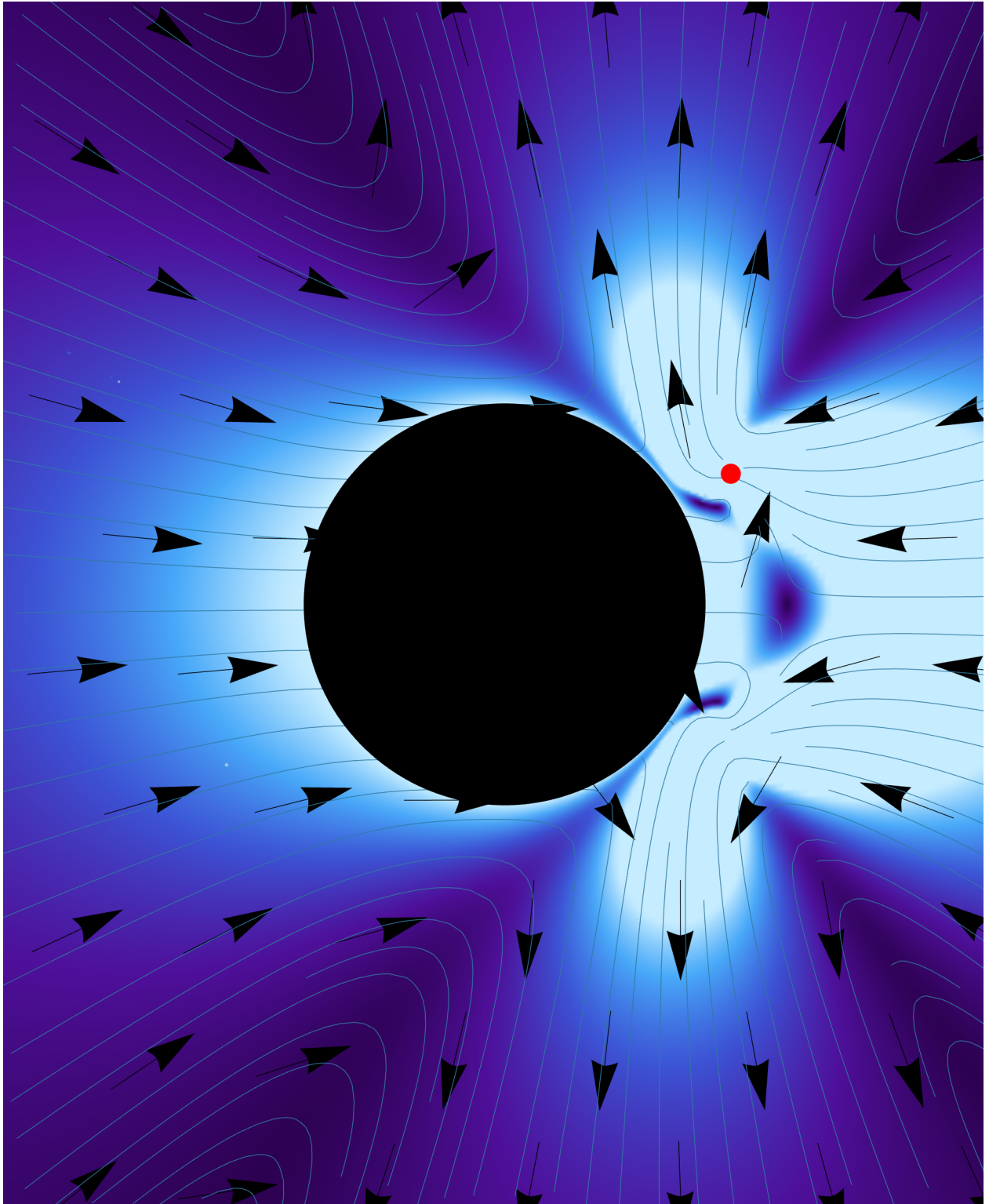
(*Plot flow in lab frame*)

```

In[46]:= mm = 3; mmoff = -0.5; rl = 0; rr = 1;
uPlot = uSwimmerN[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]}, 1, 1, 0, 12];
uPlot = {uPlot[[3]], uPlot[[1]]};
FlowALabPlot = Show[DensityPlot[Evaluate[N[Norm[uPlot]]],
  {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, PlotRange → {-10 000 rl, 10 000 rr},
  ColorFunctionScaling → False, ColorFunction →
    (ColorData["DeepSeaColors"] [ (If[# > rr, rr, If[# < -rl, -rl, #]] + rl) / (rl + rr) ] &),
  PlotPoints → 200],
  VectorPlot[uPlot, {z, -mm - mmoff, mm - mmoff},
    {x, -mm, mm}, VectorColorFunction → None, VectorStyle → Black,
    VectorPoints → 8, VectorScaling → None, VectorSizes → 1],
  (*StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, StreamColorFunction → None,
    StreamStyle → Black, StreamPoints → {Automatic, 0.8}, StreamScale → Large], *)
  StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm},
    StreamColorFunction → None, StreamPoints → {Automatic, 0.2}, StreamScale → None],
  Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
  PlotRangePadding → None, ImageSize → 800]
LegendToSave = Show[DensityPlot[x / 5, {x, 0, 5}, {y, 0, 1}, AspectRatio → Automatic,
  ColorFunctionScaling → False,
  ColorFunction → (ColorData["DeepSeaColors"] [#] &), PlotPoints → 20],
  Frame → False, PlotRangePadding → None]

```

Out[49]=



Out[50]=



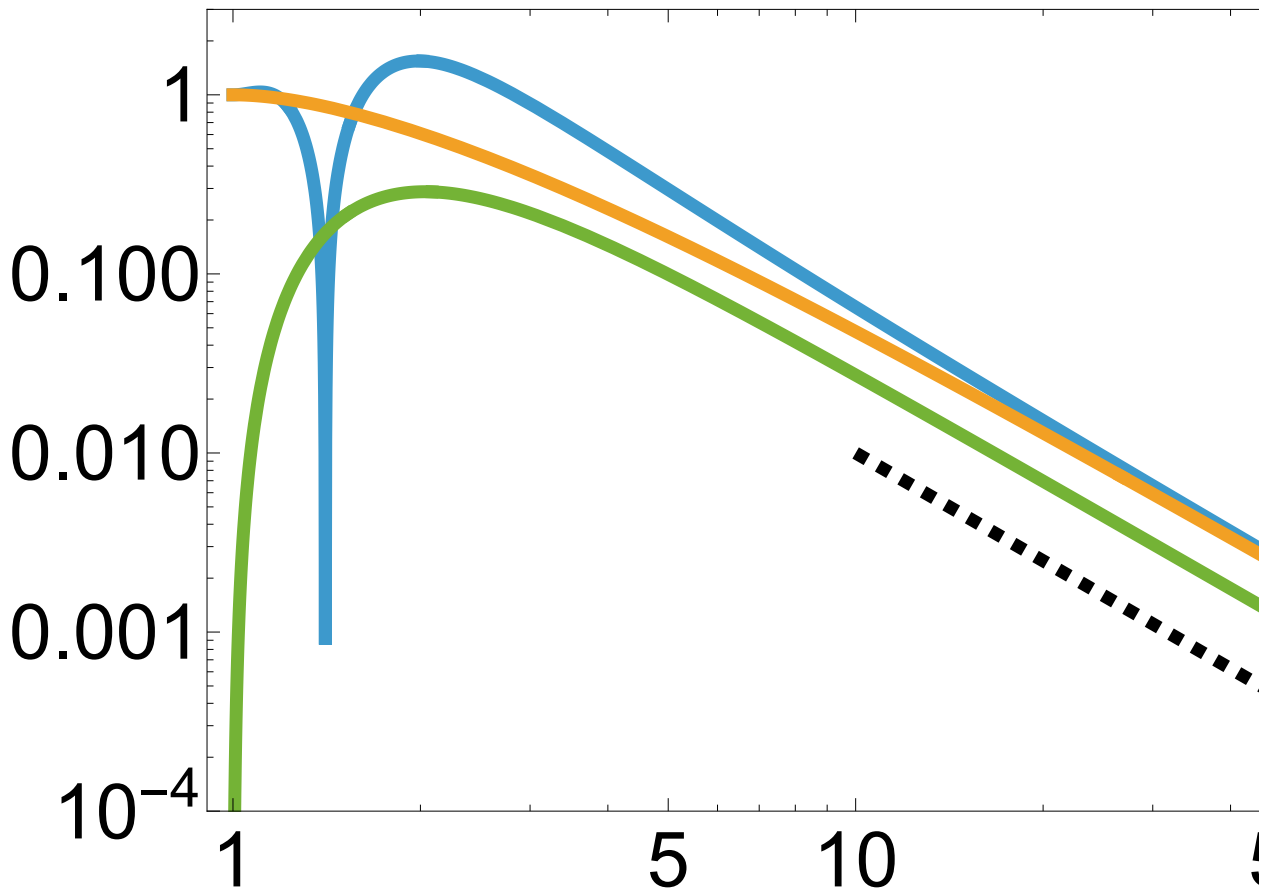
```
In[51]:= SetDirectory[NotebookDirectory[]];  
Export["FlowALab.jpg", FlowALabPlot];  
Export["FlowALegend.jpg", LegendToSave];  
NotebookSave[];
```

```

In[55]:= uPlot = uSwimmerN[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]}, 1, 1, 0, 12];
ScalePlotA = Show[
  LogLogPlot[Evaluate[
    {Norm[uPlot[[3]] /. {x → 0, z → r},
      Norm[uPlot[[3]] /. {x → 0, z → -r},
      Norm[uPlot[[1]] /. {z → 0, x → r}]}],
    {r, 1, 100}, Frame → True, PlotRange → {10-4, 3},
    PlotStyle → Thickness[0.01], AspectRatio → GoldenRatio-1],
  LogLogPlot[1/z2, {z, 10, 100},
    PlotStyle → {Thickness[0.01], Dashing[0.01], Black}],
  LabelStyle → {FontSize → 40, FontFamily → "Arial"},
  ImageSize → 800 {1, GoldenRatio-1}]

```

Out[56]=



```

In[57]:= SetDirectory[NotebookDirectory[]];
Export["ScalePlotA.jpg", ScalePlotA];
NotebookSave[];

```

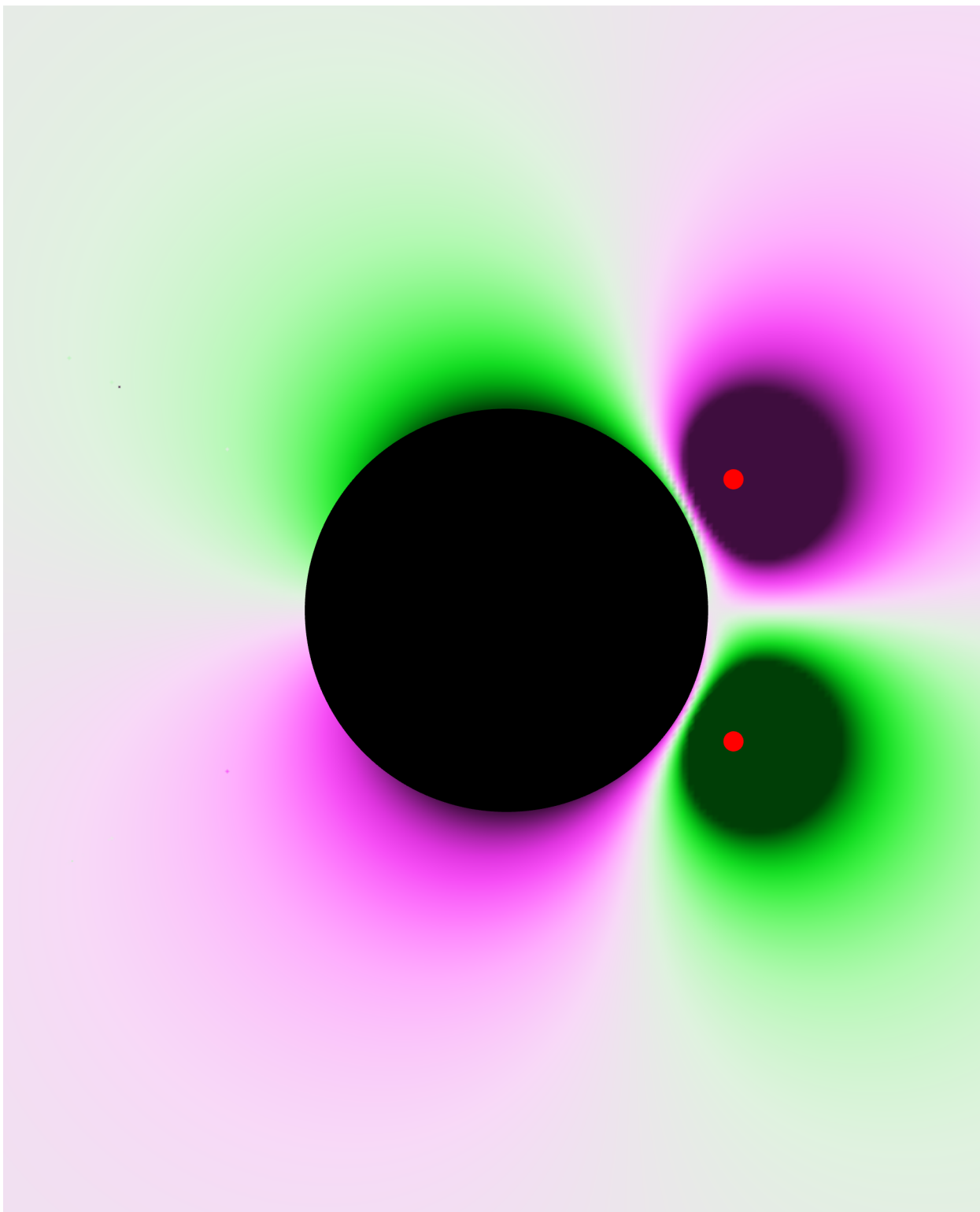
(*Spinner*)

```

In[60]:= mm = 3; mmoff = -0.5; rl = -1.; rr = 1.;
uPlot = uSwimmerN[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]}, 1, 0, 1, 12];
uPlot = -uPlot[[2]];
FlowBlabPlot = Show[DensityPlot[Evaluate[N[uPlot]],
    {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, PlotRange → {-10 000, 10 000},
    ColorFunctionScaling → False, ColorFunction →
      (ColorData["GreenPinkTones"] [ (If[# > rr, rr, If[# < rl, rl, #]] - rl) / (rr - rl) ] &),
    PlotPoints → 200],
  Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], -Sin[θvar]}, 0.05]}],
  Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
  PlotRangePadding → None, ImageSize → 800]
LegendToSave = Show[DensityPlot[x/5, {x, 0, 5}, {y, 0, 1}, AspectRatio → Automatic,
  ColorFunctionScaling → False,
  ColorFunction → (ColorData["GreenPinkTones"] [#] &), PlotPoints → 20],
  Frame → False, PlotRangePadding → None]

```

Out[63]=



Out[64]=



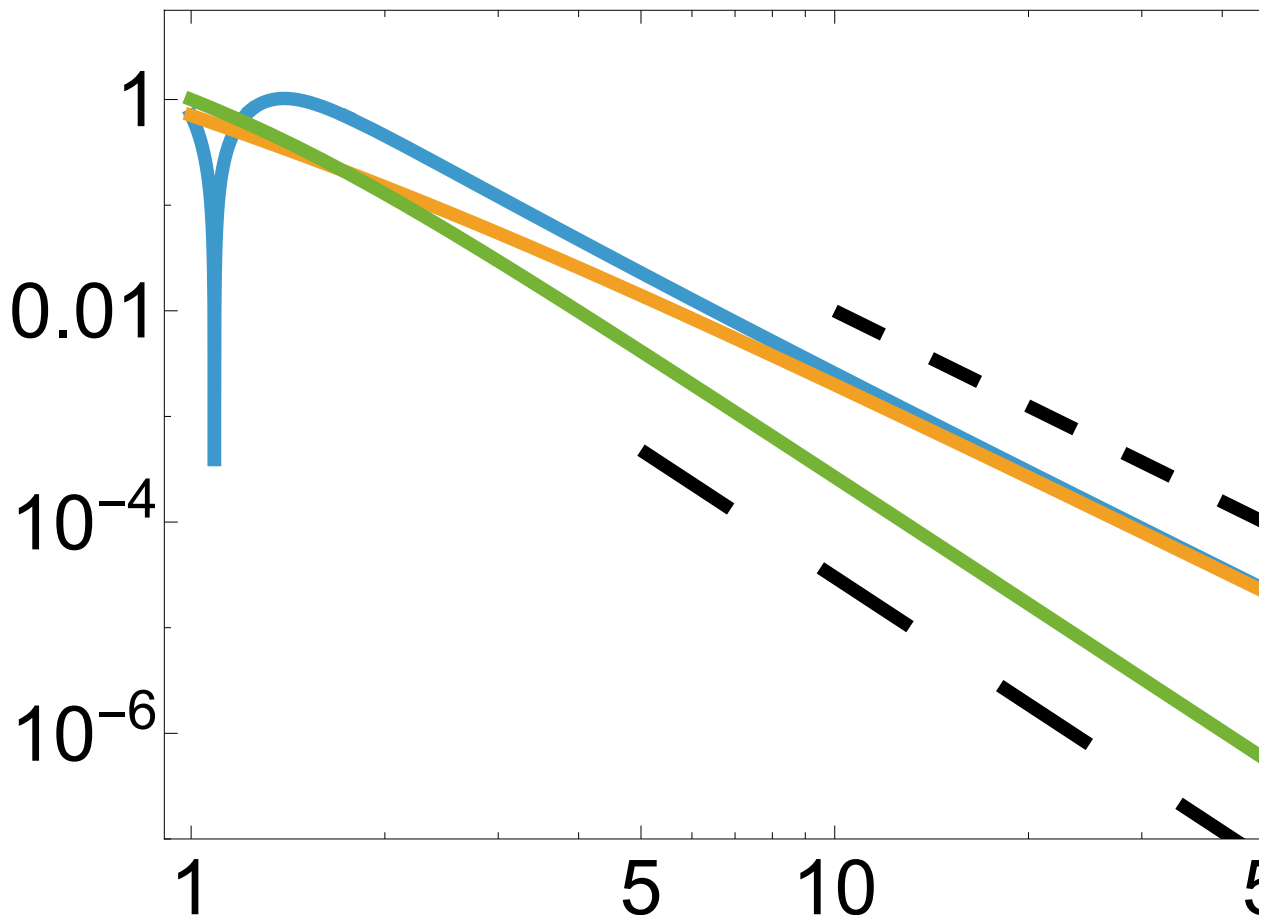
```
In[65]:= SetDirectory[NotebookDirectory[]];  
Export["FlowBLab.jpg", FlowBLabPlot];  
Export["FlowBLegend.jpg", LegendToSave];  
NotebookSave[];
```

```

In[69]:= uPlot = uSwimmerN[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]}, 1, 0, 1, 12];
ScalePlotB = Show[
  LogLogPlot[Evaluate[
    {Norm[uPlot[[2]] /. {x → r Sin[π/4], z → r Cos[π/4]},
     Norm[uPlot[[2]] /. {x → r Sin[3 π/4], z → r Cos[3 π/4]},
     Norm[uPlot[[2]] /. {z → 0, x → r}}],
    {r, 1, 100}, Frame → True, PlotRange → {10-7, 7},
    PlotStyle → Thickness[0.01], AspectRatio → GoldenRatio-1],
  LogLogPlot[10/z3, {z, 10, 100},
    PlotStyle → {Thickness[0.01], Dashing[0.04], Black}],
  LogLogPlot[0.3/z4, {z, 5, 100},
    PlotStyle → {Thickness[0.01], Dashing[0.08], Black}],
  LabelStyle → {FontSize → 40, FontFamily → "Arial"},
  ImageSize → 800 {1, GoldenRatio-1}]

```

Out[70]=



```

In[71]:= SetDirectory[NotebookDirectory[]];
Export["ScalePlotB.jpg", ScalePlotB];
NotebookSave[];

```

(*Didinium*)

```

In[74]:= uSwimmerNM[{x_, y_, z_}, {xs_, ys_, zs_}, {zf_, pf_},
  {zf2_, pf2_}, A_, V_, Ω_, NN_Integer, MM_Integer] := Module[
  {RXX = {x, y, z} - {xs, ys, zs},
  Fz =
    - 
$$\frac{1}{\left(1 - \frac{A^3 (-2 zf^2 + pf^2) + 3 A (zf^2 + pf^2) (2 zf^2 + pf^2)}{4 (zf^2 + pf^2)^{5/2}}\right) NN + \left(1 - \frac{A^3 (-2 zf2^2 + pf2^2) + 3 A (zf2^2 + pf2^2) (2 zf2^2 + pf2^2)}{4 (zf2^2 + pf2^2)^{5/2}}\right) MM}$$

    
$$\frac{6 A V}{8},$$

  Fφ = 
$$\frac{1}{pf \left(1 - \left(\frac{A^2}{zf^2 + pf^2}\right)^{3/2}\right) NN + pf2 \left(1 - \left(\frac{A^2}{zf2^2 + pf2^2}\right)^{3/2}\right) MM} A^3 \Omega},
  (*Print[Fz];
  Print[Fφ];
  Print[ArcTan[Fφ/Fz] 180./π];*)
  
$$\frac{3 A V}{4} \left( \text{Gbulk}[\{x, y, z\}, \{xs, ys, zs\}] \llbracket \text{All}, 3 \rrbracket + \frac{A^2}{6} \text{GbulkLap}[\{x, y, z\}, \{xs, ys, zs\}] \right)$$

  + 
$$\frac{A^3}{\text{Norm}[RXX]^3} \text{Cross}[\{0, 0, \Omega\}, RXX]$$

  + Sum[
$$\left( \text{Gbulk}[\{x, y, z\}, \{xs, ys, zs\}] + \left\{ pf \cos\left[2 \pi \frac{i-1}{NN}\right], pf \sin\left[2 \pi \frac{i-1}{NN}\right], zf \right\} \right) +$$


$$\text{Gsphere}\left[\{x, y, z\}, \left\{ pf \cos\left[2 \pi \frac{i-1}{NN}\right], pf \sin\left[2 \pi \frac{i-1}{NN}\right], zf \right\}, \{xs, ys, zs\}, A \right].$$


$$\left\{ F\phi \sin\left[2 \pi \frac{i-1}{NN}\right], -F\phi \cos\left[2 \pi \frac{i-1}{NN}\right], Fz \right\}, \{i, 1, NN\}]$$

  + Sum[
$$\left( \text{Gbulk}[\{x, y, z\}, \{xs, ys, zs\}] + \left\{ pf2 \cos\left[2 \pi \frac{i-1}{MM}\right], pf2 \sin\left[2 \pi \frac{i-1}{MM}\right], zf2 \right\} \right) +$$


$$\text{Gsphere}\left[\{x, y, z\}, \left\{ pf2 \cos\left[2 \pi \frac{i-1}{MM}\right], pf2 \sin\left[2 \pi \frac{i-1}{MM}\right], zf2 \right\}, \{xs, ys, zs\}, A \right].$$


$$\left\{ F\phi \sin\left[2 \pi \frac{i-1}{MM}\right], -F\phi \cos\left[2 \pi \frac{i-1}{MM}\right], Fz \right\}, \{i, 1, MM\}]$$

  ]$$

```

(*Lab frame*)

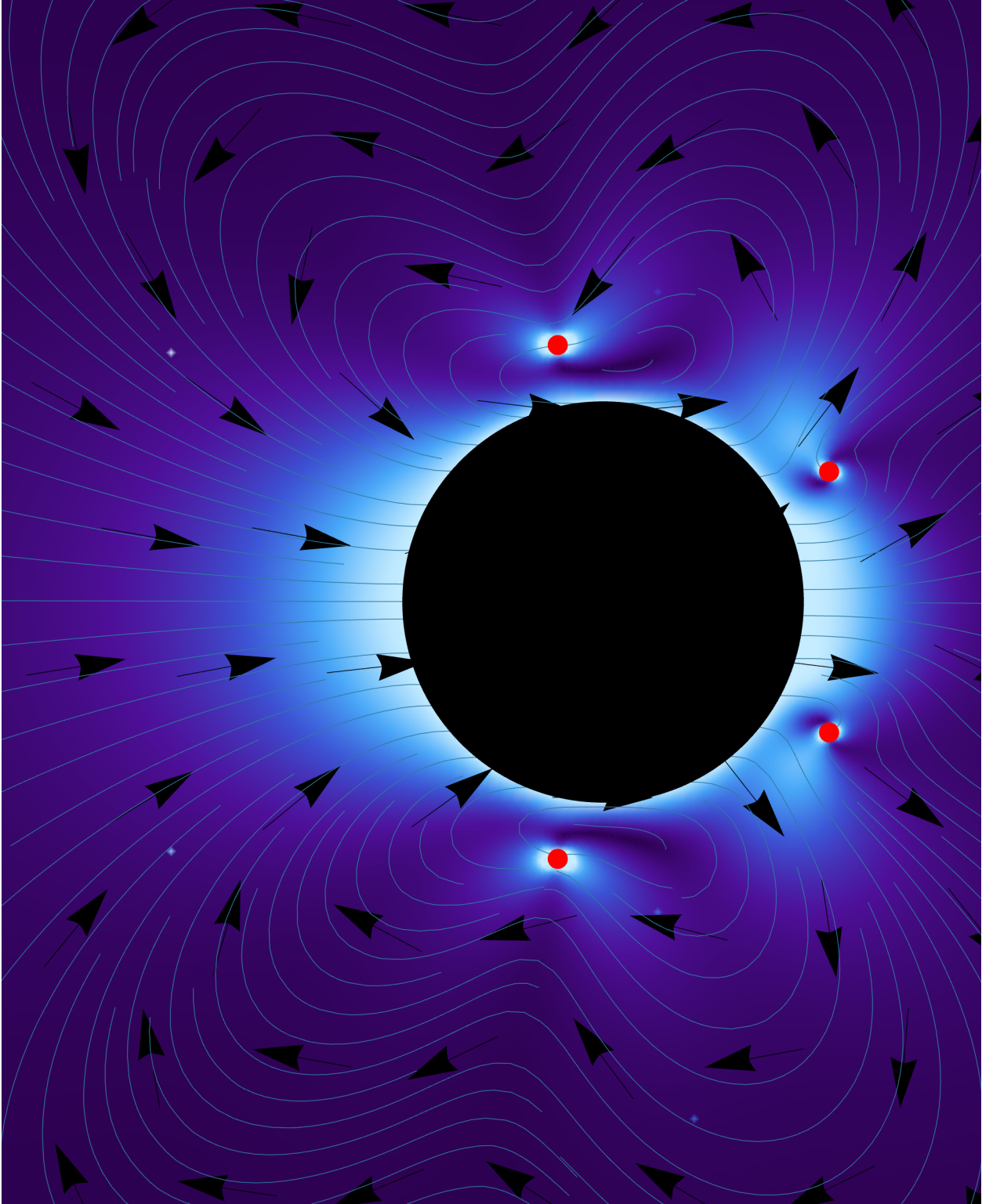
```

In[75]:= mm = 3; mmoff = 0; rl = 0; rr = 1;
{ρvar, ρ2var, θvar 180. / π, θ2var 180. / π}
uPlot = uSwimmerNM[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]},
  ρ2var {Cos[θ2var], Sin[θ2var]}, 1., 1., 0 × 0.4438, 24, 48];
uPlot = {uPlot[[3]], uPlot[[1]]};
FlowCLabPlot = Show[DensityPlot[Evaluate[N[Norm[uPlot]]],
  {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, PlotRange → {-1000, 1000},
  ColorFunctionScaling → False, ColorFunction →
    (ColorData["DeepSeaColors"] [ (If[# > rr, rr, If[# < rl, rl, #]] - rl) /
      (rr - rl) ] &),
  PlotPoints → 100],
  VectorPlot[uPlot, {z, -mm - mmoff, mm - mmoff},
    {x, -mm, mm}, VectorColorFunction → None, VectorStyle → Black,
    VectorPoints → 8, VectorScaling → None, VectorSizes → 1],
  (*StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, StreamColorFunction → None,
    StreamStyle → Black, StreamPoints → {Automatic, 0.8}, StreamScale → Large], *)
  StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm},
    StreamColorFunction → None, StreamPoints → {Automatic, 0.2}, StreamScale → None],
  Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], -Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], Sin[θ2var]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], -Sin[θ2var]}, 0.05]}],
  Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
  PlotRangePadding → None, ImageSize → 800]
LegendToSave = Show[DensityPlot[x / 5, {x, 0, 5}, {y, 0, 1}, AspectRatio → Automatic,
  ColorFunctionScaling → False,
  ColorFunction → (ColorData["DeepSeaColors"] [#] &), PlotPoints → 20],
  Frame → False, PlotRangePadding → None]

Out[76]= {1.3, 1.3, 30., 100.}

```

Out[79]=



Out[80]=



```

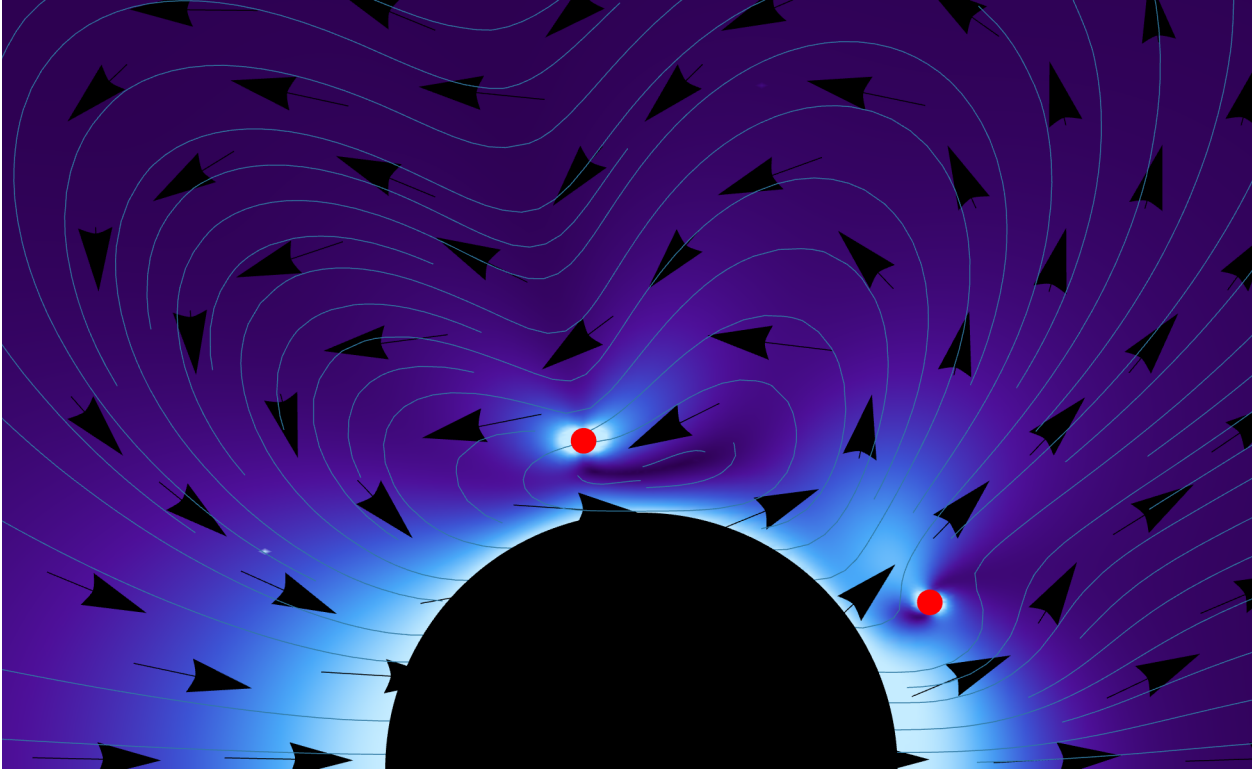
In[81]:= mm = 3; mmoff = -0.5; rl = 0; rr = 1;
{ρvar, ρ2var, θvar 180. / π, θ2var 180. / π}
uPlot = uSwimmerNM[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]},
  ρ2var {Cos[θ2var], Sin[θ2var]}, 1., 1., 0 × 0.4438, 24, 48];
uPlot = {uPlot[[3]], uPlot[[1]]};
FlowCLabPlot = Show[DensityPlot[Evaluate[N[Norm[uPlot]]],
  {z, -mm - mmoff, mm - mmoff}, {x, 0, mm}, PlotRange → {-1000, 1000},
  ColorFunctionScaling → False, ColorFunction →
    (ColorData["DeepSeaColors"] [ (If[# > rr, rr, If[# < rl, rl, #]] - rl) / (rr - rl) ] &),
  PlotPoints → 100, AspectRatio → Automatic],
  VectorPlot[uPlot, {z, -mm - mmoff, mm - mmoff},
    {x, 0, mm}, VectorColorFunction → None, VectorStyle → Black,
    VectorPoints → 8, VectorScaling → None, VectorSizes → 1],
  (*StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, -mm, mm}, StreamColorFunction → None,
    StreamStyle → Black, StreamPoints → {Automatic, 0.8}, StreamScale → Large], *)
  StreamPlot[uPlot, {z, -mm - mmoff, mm - mmoff}, {x, 0, mm},
    StreamColorFunction → None, StreamPoints → {Automatic, 0.2}, StreamScale → None],
  Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], Sin[θ2var]}, 0.05]}],
  Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
  PlotRangePadding → None, ImageSize → 800 {1, 0.5}]

```

Out[82]=

```
{1.3, 1.3, 30., 100.}
```

Out[85]=



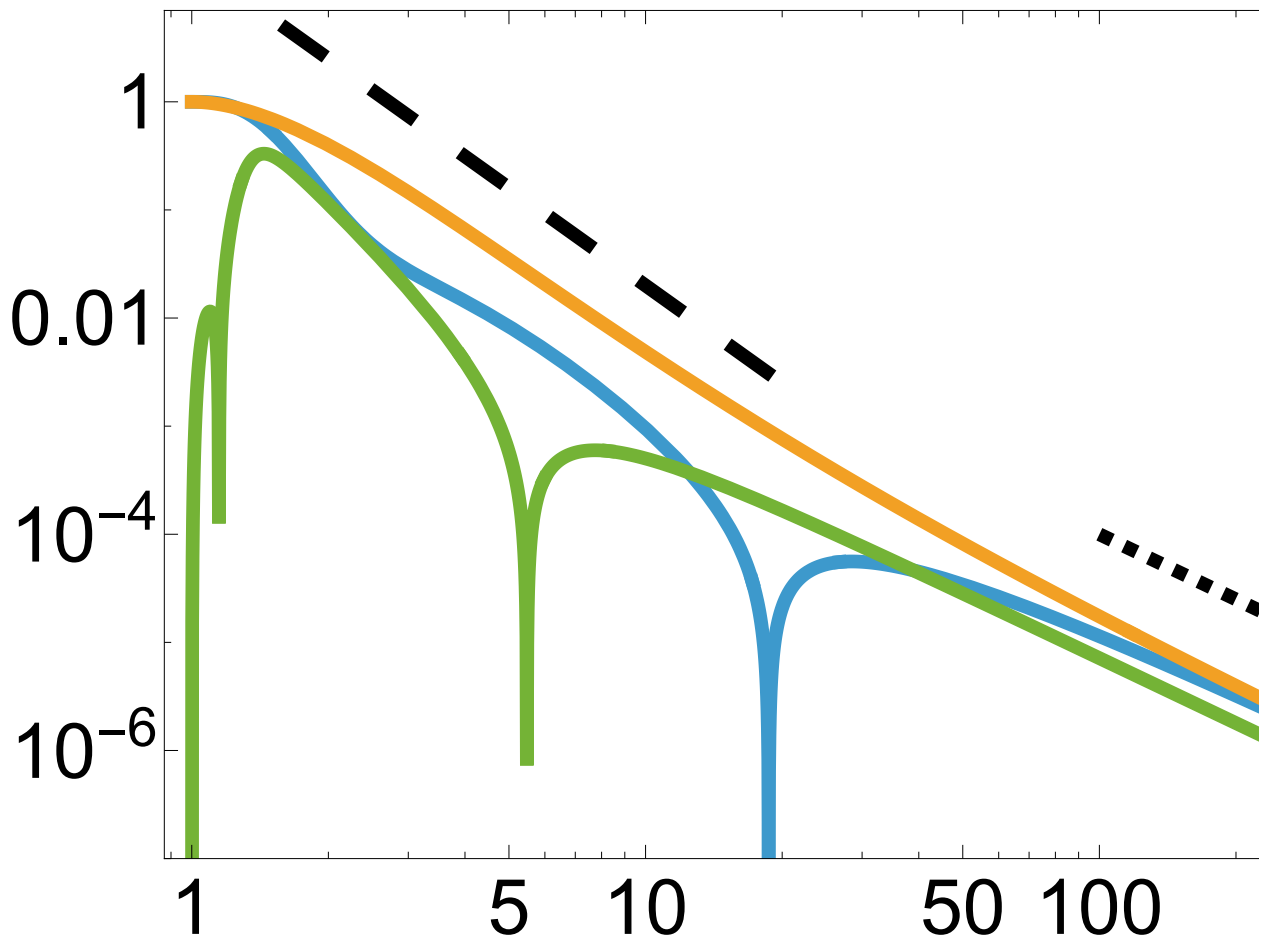
```
In[86]:= SetDirectory[NotebookDirectory[]];  
Export["FlowCLab.jpg", FlowCLabPlot];  
NotebookSave[];
```

```

In[89]:= uPlot = uSwimmerNM[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]},
  ρ2var {Cos[θ2var], Sin[θ2var]}, 1., 1., 0×0.4438, 24, 48];
ScalePlotC = Show[
  (*LogLogPlot[Norm[uPlot[[3]] /. {x→0}, {z, 1, 800}, Frame→True, PlotRange→{10-7, 1}], *)
  LogLogPlot[Evaluate[
    {Norm[uPlot[[3]] /. {x→0, z→r},
      Norm[uPlot[[3]] /. {x→0, z→-r},
      Norm[uPlot[[1]] /. {z→0, x→r}]}],
    {r, 1, 800}, Frame→True, PlotRange→{10-7, 7},
    PlotStyle→Thickness[0.01], AspectRatio→GoldenRatio-1],
  LogLogPlot[1/z2, {z, 100, 1000},
    PlotStyle→{Thickness[0.01], Dashing[0.01], Black}],
  LogLogPlot[20/z3, {z, 1, 20},
    PlotStyle→{Thickness[0.01], Dashing[0.04], Black}]
  , LabelStyle→{FontSize→40, FontFamily→"Arial"},
  ImageSize→800 {1, GoldenRatio-1}]

```

Out[90]=



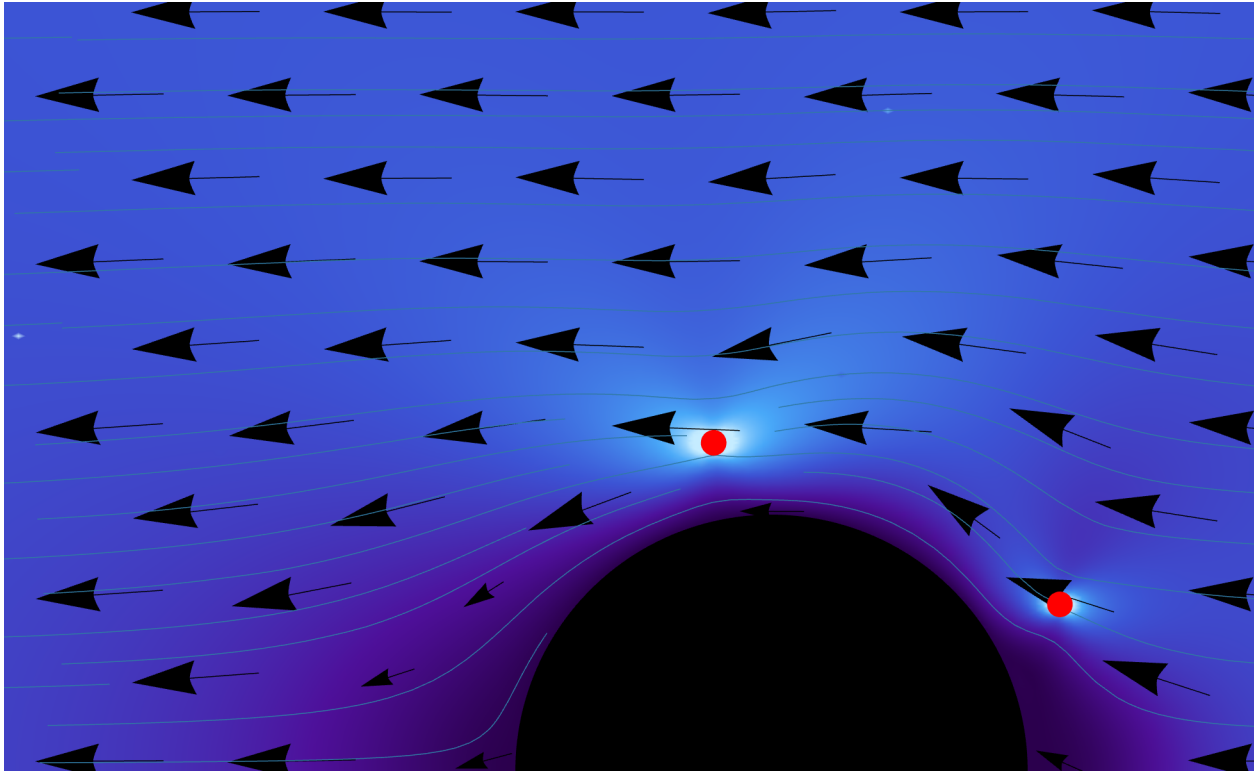
```
In[91]:= SetDirectory[NotebookDirectory[]];
Export["ScalePlotC.jpg", ScalePlotC];
NotebookSave[];
```

(*Rest frame*)

```
In[94]:= mm = 3; mmoff = 0; rl = 0; rr = 2;
{ρvar, ρ2var, θvar 180. / π, θ2var 180. / π}
uPlot = uSwimmerNM[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]},
  ρ2var {Cos[θ2var], Sin[θ2var]}, 1., 1., 0.4438, 24, 48];
uPlot = {uPlot[[3]], uPlot[[1]]};
FlowCrestPlot = Show[DensityPlot[Evaluate[N[Norm[{-1, 0} + uPlot]]],
  {z, -mm - mmoff, mm - mmoff}, {x, 0, mm}, PlotRange → {-1000, 1000},
  ColorFunctionScaling → False, ColorFunction →
    (ColorData["DeepSeaColors"] [ (If[# > rr, rr, If[# < rl, rl, #]] - rl) /
      (rr - rl) ) & ],
  PlotPoints → 100, AspectRatio → Automatic],
  VectorPlot[{-1, 0} + uPlot, {z, -mm - mmoff, mm - mmoff},
    {x, 0, mm}, VectorColorFunction → None, VectorStyle → Black,
    VectorPoints → 8, VectorScaling → None, VectorSizes → 1],
  (*StreamPlot[{-1, 0} + uPlot, {z, -mm - mmoff, mm - mmoff},
    {x, -mm, mm}, StreamColorFunction → None, StreamStyle → Black,
    StreamPoints → {Automatic, 0.8}, StreamScale → Large], *)
  StreamPlot[{-1, 0} + uPlot, {z, -mm - mmoff, mm - mmoff}, {x, 0, mm},
    StreamColorFunction → None, StreamPoints → {Automatic, 0.2}, StreamScale → None],
  Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], -Sin[θvar]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], Sin[θ2var]}, 0.05]}],
  Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], -Sin[θ2var]}, 0.05]}],
  Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
  PlotRangePadding → None, ImageSize → 800 {1, 0.5}]

Out[95]= {1.3, 1.3, 30., 100.}
```

Out[98]=



```
In[99]:= SetDirectory[NotebookDirectory[]];  
Export["FlowCrest.jpg", FlowCrestPlot];  
NotebookSave[];
```

(*Spinning*)

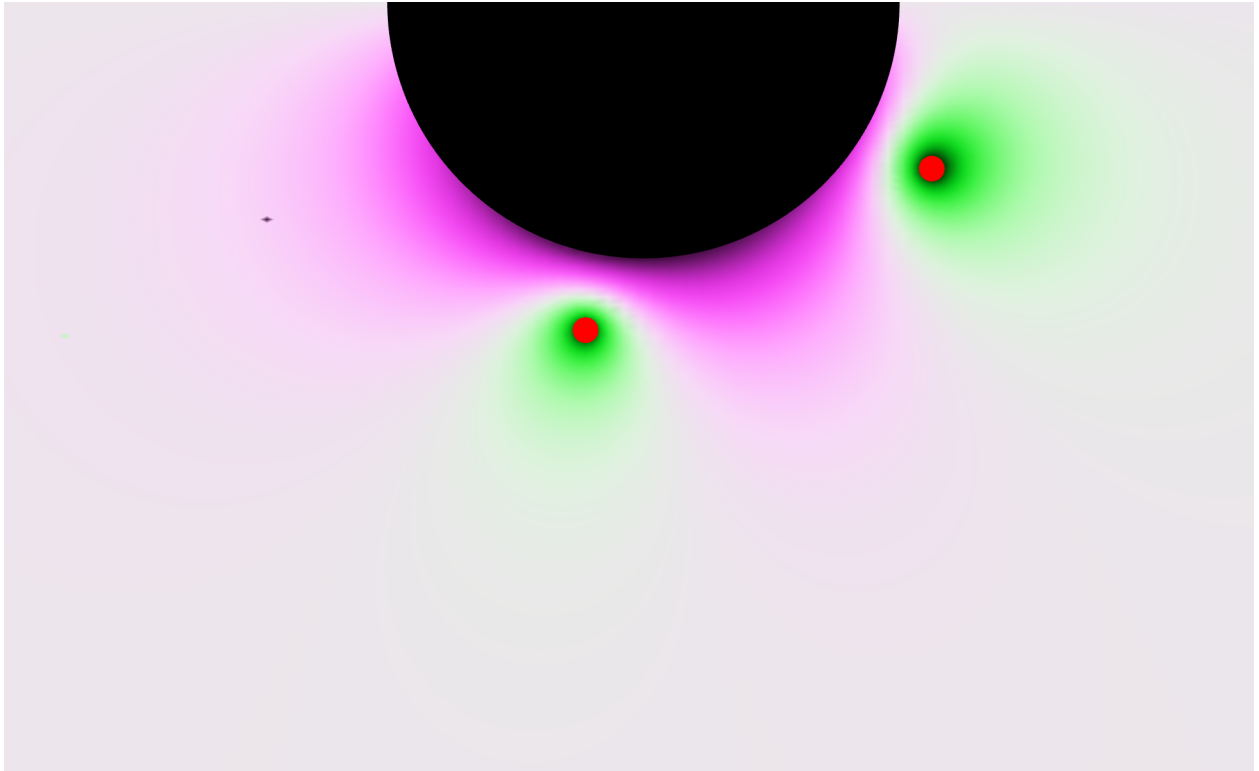
In[102]:=

```
mm = 3; mmoff = -0.5; rl = -1; rr = 1;
{ρvar, ρ2var, θvar 180. / π, θ2var 180. / π}
uPlot = uSwimmerNM[{x, 0, z}, {0, 0, 0}, ρvar {Cos[θvar], Sin[θvar]},
  ρ2var {Cos[θ2var], Sin[θ2var]}, 1., 1., 0.4438, 24, 48];
uPlot = - $\frac{1}{0.4438}$  uPlot[[2]];
FlowCrotPlot = Show[DensityPlot[Evaluate[N[uPlot]],
  {z, -mm - mmoff, mm - mmoff}, {x, -mm, 0}, PlotRange → {-10 000, 10 000},
  ColorFunctionScaling → False, ColorFunction →
    (ColorData["GreenPinkTones"] [  $\left( \frac{\text{If}[\# > rr, rr, \text{If}[\# < rl, rl, \#]] - rl}{(rr - rl)} \right)$  ] &),
  PlotPoints → 100, AspectRatio → Automatic],
Graphics[{Thick, Black, Disk[{0, 0}, 1]}],
Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], Sin[θvar]}, 0.05]}],
Graphics[{Thick, Red, Disk[ρvar {Cos[θvar], -Sin[θvar]}, 0.05]}],
Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], Sin[θ2var]}, 0.05]}],
Graphics[{Thick, Red, Disk[ρ2var {Cos[θ2var], -Sin[θ2var]}, 0.05]}],
Frame → False, LabelStyle → {FontSize → 20, FontFamily → "Arial"},
PlotRangePadding → None, ImageSize → 800 {1, 0.5}]
```

Out[103]=

```
{1.3, 1.3, 30., 100.}
```

Out[106]=



In[107]:=

```
SetDirectory[NotebookDirectory[]];  
Export["FlowCRot.jpg", FlowCRotPlot];  
NotebookSave[];
```